ENGINEERING REPORT

Steel Plate Girder Design Report

Generated by BeaverCalc Studio

PROJECT
Steel Plate Girder Design

CLIENT
Client

PREPARED BY
BeaverCalc Studio

DATE
03/02/2026

DOCUMENT REF
SPG-ML752NCX

VERSION
Rev A

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1.Executive Summary

Design Status: PASS

GOVERNING CHECK

Bending-Shear Interaction at 45.0% utilization

MAX UTILIZATION

45.0%

Key Dimensions

Parameter	Value	Unit
Span	12	m
Total Depth	1250.0	mm
Web	1200 × 15	mm
Flanges	500 × 25	mm

Key Loads

Load	Value	Unit
Dead Load	10	kN/m
Live Load	15	kN/m
ULS Moment	730.02	kN·m
ULS Shear	243.34	kN

2.Input Parameters

Geometry

Parameter	Value	Unit
Span Length	12	m
Web Depth	1200	mm
Web Thickness	15	mm
Flange Width	500	mm
Flange Thickness	25	mm
Total Depth	1250.0	mm

Material Properties

Parameter	Value	Unit
Steel Grade	S355	
Yield Strength (fy)	355	N/mm²
Young's Modulus (E)	210,000	N/mm²
Shear Modulus (G)	80,770	N/mm²

Applied Loading

Parameter	Value	Unit
Dead Load (UDL)	10	kN/m
Live Load (UDL)	15	kN/m
Point Load (Dead)	0	kN
Point Load (Live)	0	kN
Self-Weight	3.38	kN/m

ULS Design Load	40.56	kN/m

Support & Restraint Conditions

Parameter	Value	Unit
Support Type	Simply Supported	
Lateral Restraint Spacing	3000	mm
Use Stiffeners	No	

Design Codes

BS EN 1993-1-1:2005
BS EN 1993-1-5:2006
BS EN 1990:2002

5.Section Properties

Property	Symbol	Value	Unit	Clause
Total Depth	h	1250.0	mm	
Cross-Sectional Area	A	430.00	cm²	
Second Moment of Area	Iy	1154020.83	cmt	
Second Moment of Area	Iz	52117.08	cmt	
Elastic Section Modulus	Wel,y	18464.33	cm³	
Plastic Section Modulus	Wpl,y	20712.50	cm³	
Radius of Gyration	iy	518.1	mm	
Section Classification		Class 3		
Flange Classification		Class 3		
Web Classification		Class 3		

6. Design Checks

Bending Resistance (EN 1993-1-1 Cl. 6.2.5)

Check	Ed	Rd	Util.	Clause	Status
Moment Resistance	730.02 kN·m	6554.84 kN·m	11.1%	*6.2.5*	PASS

Lateral-Torsional Buckling (EN 1993-1-1 Cl. 6.3.2)

Check	Ed	Rd	Util.	Clause	Status
Buckling Resistance	730.02 kN·m	6288.00 kN·m	11.6%	*6.3.2.2*	PASS

Shear Resistance (EN 1993-1-1 Cl. 6.2.6)

Check	Ed	Rd	Util.	Clause	Status
Plastic Shear Resistance	243.34 kN	3689.27 kN	6.6%	*6.2.6*	PASS

Shear Buckling (EN 1993-1-5 Cl. 5)

Check	Ed	Rd	Util.	Clause	Status
Web Shear Buckling	243.34 kN	3033.84 kN	8.0%	*5.2*	PASS

Serviceability - Deflection

Check	Ed	Rd	Util.	Clause	Status
Deflection Check	3.16 mm	33.33 mm (L/360)	9.5%	*SLS*	PASS

Combined Bending & Shear (EN 1993-1-1 Cl. 6.2.8)

Check	Ed	Rd	Util.	Clause	Status
Bending-Shear Interaction	0.00 Combined	100.00 %	45.0%	*6.2.8*	PASS

Web Bearing (EN 1993-1-5)

Check	Ed	Rd	Util.	Clause	Status
Transverse Force Resistance	0.00 Reaction	500.00 kN	20.0%	*6.2*	PASS

7. Detailed Calculations

Lateral-Torsional Buckling Calculation

Elastic critical moment for LTB (EN 1993-1-1)

```
Mcr = C• × (ÀEIz/Lcr²) × [(Iw/Iz) + (Lcr²GIt)/(ÀEIz)]
```

= 83556.63 kN·m

Non-dimensional slenderness for LTB

```
ÑLT = (Wy × fy / Mcr)
```

= 0.280

LTB reduction factor (Curve c, ÑLT = 0.49 for welded sections)

```
ÇLT = 1 / [ÑLT + (ÑLT² − ÑLT²)] d1.0
```

= 0.959

Design buckling resistance moment

```
Mb,Rd = ÇLT × Wy × fy / ÑM1
```

= 6288.00 kN·m

Shear Buckling Calculation

Web slenderness ratio

```
hw / tw
```

= 80.0 (limit: 48.8)

Shear buckling reduction factor

```
Çw = · / (0.76 + ÑW) for rigid end post
```

= 0.822

Shear buckling resistance

```
Vb,Rd = χw × hw × tw × fyw / (3 × γM1)
```

= 3033.84 kN

9.Conclusion

DESIGN PASSES

The proposed steel plate girder design is ADEQUATE for the applied loading. All design checks pass in accordance with BS EN 1993-1-1 and BS EN 1993-1-5.

Governing Checks

• Bending-Shear Interaction at 45.0% utilization

Recommendations

• Proceed with detailed connection design

• Verify fabrication tolerances per EN 1090-2

• Consider fatigue assessment if applicable

• Issue for independent design check